

Innovation as prospective-but-not-retrospective surprise: Kullback–Leibler novelty scores on 2.0 million USPTO trademark filings

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Abstract

We operationalize firm-level innovation by applying Simon DeDeo’s prospective/retrospective surprise framework to the goods-and-services descriptions of every U.S. trademark filed since 1985 in two NICE classes: Class 9 (Software & Electronics) and Class 32 (Beer & Soft Drinks). For each filing at calendar year t we compute the Kullback–Leibler divergence of its term distribution from a 5-year look-behind reference (*prospective* surprise) and from a 5-year look-ahead reference (*retrospective* surprise). The contrast $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ is positive when a filing was unusual at the time but normal in hindsight — the empirical signature of a firm that introduced a category later imitated — and negative when the past predicts the filing better than the future does.

The score recovers the risk–reward asymmetry that the literature on pioneers and fast-followers predicts. In a generalized linear logit on 1,146,415 Class-9 filings, a one-standard-deviation increase in ΔKL *lowers* the probability of reaching registration by 0.042 in log-odds (SE = 0.002) but *raises* the probability of being a currently-live mark in 2026 by 0.038 (SE = 0.002). Linking the trademark scores to SEC EDGAR financials for 1,408 matched publicly-traded firms gives the same pattern at the firm-year level: gross margin rises with ΔKL at +2.9 percentage points per σ (SE 0.7 pp). The decomposition shows that the gain comes from firms whose filings the future ends up resembling, not from firms whose filings stand alone.

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.’s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including

most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. LIQUID DEATH’s canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two stylized stories are in tension. The first, conveyed in the phrase “you can tell the pioneers by the arrows in their backs,” is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, popularized by the *Blue Ocean Strategy* literature, is that firms that create uncontested category space enjoy outsized margins, at least conditional on survival. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

We download the entire USPTO Trademark Annual Applications backfile (product TRTYRAP, ~ 12.4 GB across 91 ZIP parts as of April 2026) through the Open Data Portal API. Records are streamed: each ZIP is downloaded, parsed in memory with `lxml.iterparse`, filtered to the target NICE international class, and written incrementally to Parquet, after which the raw ZIP is deleted. From each `<case-file>` record we retain the serial number, filing date, registration date, status code, mark identification, NICE class list, US class list, registrant name (the owner at the time of filing), and goods-and-services text (case-file statements with type-code prefix `GS`). Mark-description statements (prefix `D`) are excluded because they describe the visual mark, not the business.

Our two industries are NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). Class 9 was chosen as the canonical software class with high-volume long-tailed structure; Class 32 was chosen for its tractable size and well-known innovation cycles (microbrew, energy drinks, hard seltzer, cannabis beverages). Owner names are taken verbatim from the USPTO XML; entity resolution to canonical firm identifiers is described in Section 7. Per-class summary statistics are reported in Appendix A.

For firm-level outcome data we use the SEC EDGAR Financial Statement Data Sets (FSDS), 64 quarterly XBRL releases from 2009Q1 through 2024Q4. The FSDS covers 13,127 unique CIKs after filtering to consolidated 10-K income-statement and balance-sheet tags. We extract revenue, cost of revenue, gross profit, R&D expense, operating income, and net income.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each calendar year t we form two reference distributions by aggregating term frequencies

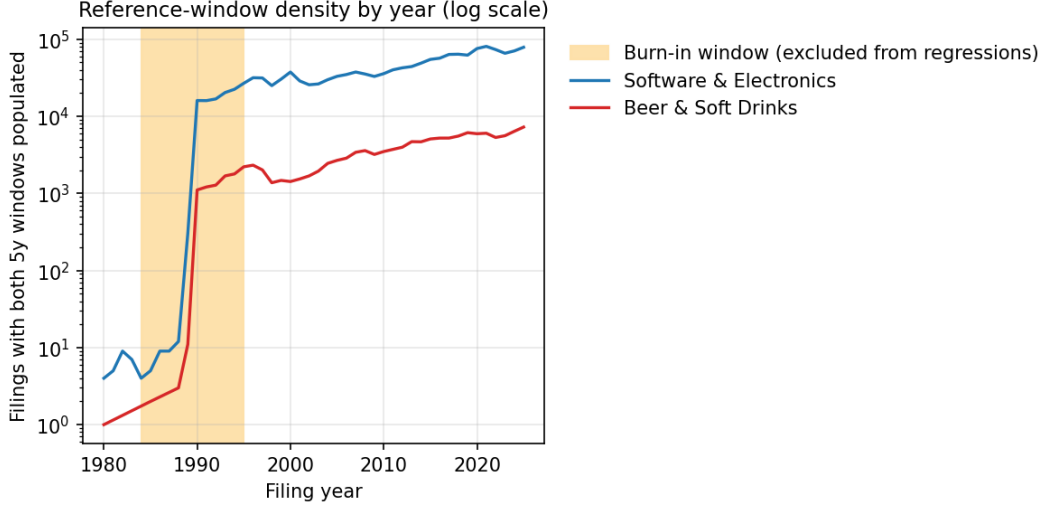


Figure 1: Filings per year with both 5-year reference windows populated. The shaded region marks years where the prospective window is sparse. For all reported regressions we restrict to filings from 1995 onward.

across neighboring years:

$$Q_t^{\text{pros}}(v) \propto \alpha + \sum_{j: \text{year}(j) \in [t-W, t]} \text{count}(v, j), \quad (1)$$

$$Q_t^{\text{retr}}(v) \propto \alpha + \sum_{j: \text{year}(j) \in (t, t+W]} \text{count}(v, j), \quad (2)$$

with window $W = 5$ years (look-behind for prospective, look-ahead for retrospective) and Laplace smoothing $\alpha = 1.0$. The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_t^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_t^{\text{retr}}(v)}.$$

The directional innovation score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$. We use a term-frequency multinomial as the v0 baseline; an LDA topic-model replication is planned but not reported here.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token **Beer**; many app developers file **downloadable software**. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

Burn-in caveat. The 5-year prospective window for filings near 1985 reaches into a period when the corpus is sparse (Figure 1). The reference Q^{pros} has high entropy, and any specific filing looks far from it. We address this by restricting all regressions to filings from 1995 onward.

4 Face validity

Figure 2 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high $\Rightarrow$ less like the past); the

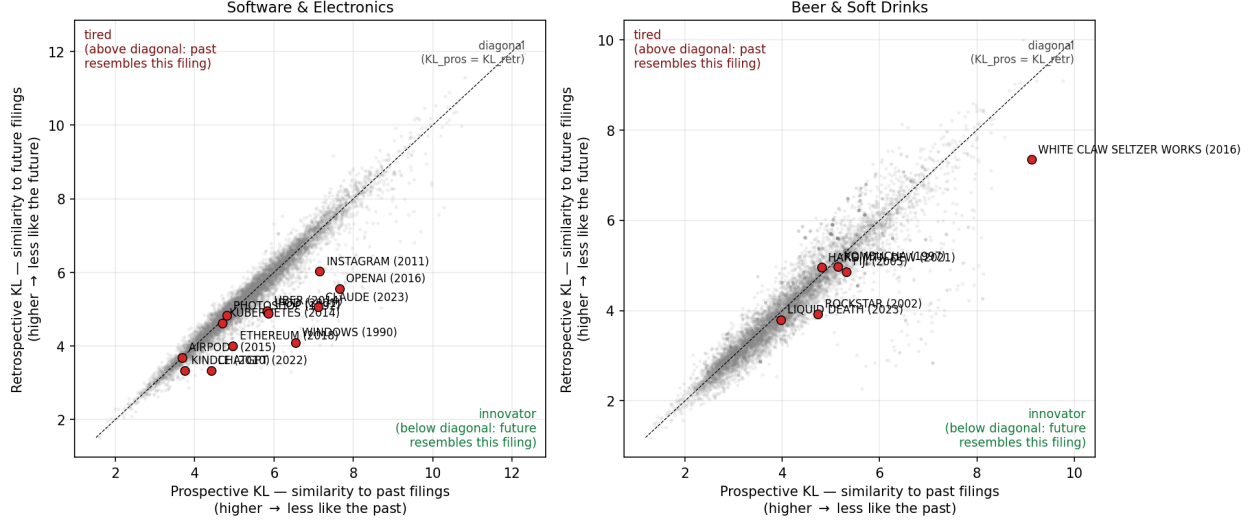


Figure 2: Per-filing prospective vs. retrospective KL surprise. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). Below-diagonal: innovator (the future resembles this filing more than the past does). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high $\Rightarrow$ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OPENAI 2016, CLAUDE 2023, INSTAGRAM 2011, WHITE CLAW SELTZER WORKS 2016, ROCKSTAR 2002) and expectation-disconfirming late-cycle marks (HARD MTN DEW 2021).

The bottled-water category illustrates a useful failure mode. The 2019 LIQUID DEATH filing for canned water describes its goods as *still water; sparkling water* — the same lexicon as established bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (SARATOGA, SMARTWATER, ESSENTIA) cluster in the same region. The method is, by construction, blind to non-textual novelty.

5 Industry time series

Figures 3 and 4 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

6 Outcome 1: Trademark survival

We derive four binary outcomes per filing from the USPTO status code.

- **Reached registration:** filing has a non-null `registration_date`. The mark cleared examination, was published for opposition, and was registered. Mean elapsed time from filing to

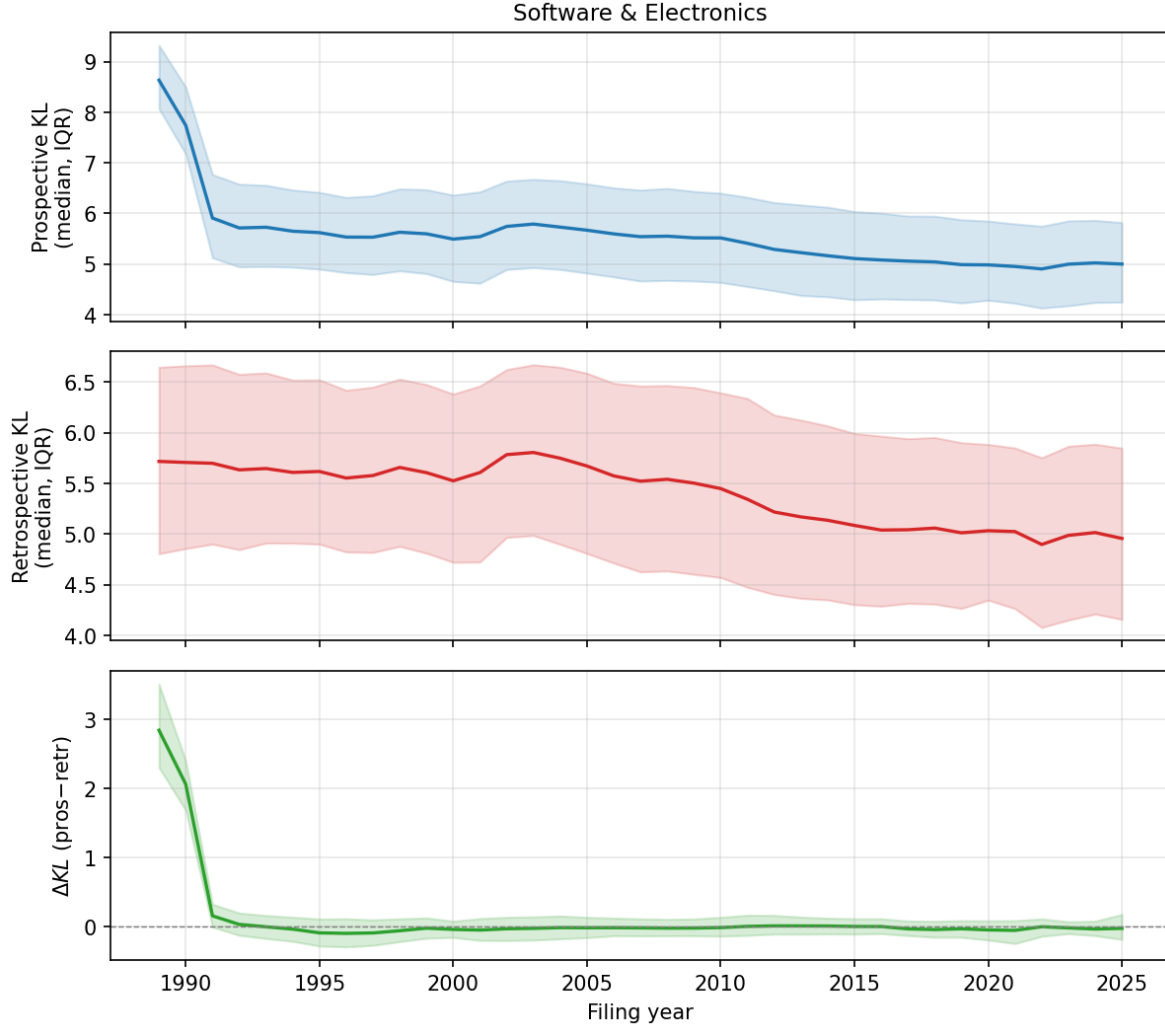


Figure 3: Software & Electronics: per-year median (line) and 25–75% range (band) of prospective KL, retrospective KL, and ΔKL . The first decade (≤ 1994) shows inflated absolute KL because the 5-year look-behind reaches into a sparse corpus.

registration is roughly 11–13 months in our corpus.

- **Survived 5 years:** reached registration, was filed at least 5 years before today (we cap at 2020 cohorts), and was not cancelled post-registration. The 5-year mark coincides with the first Section 8 maintenance filing; failure to file is the most common cause of early cancellation.
- **Survived 10 years:** reached registration, filed at least 10 years ago (we cap at 2015 cohorts), and not cancelled. The 10-year mark is the first Section 9 renewal; subsequent renewals are at 10-year intervals indefinitely.
- **Currently live (April 2026):** status code begins with 6, regardless of cohort.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2(t_i - \bar{t}) + \gamma_3(t_i - \bar{t})^2)$$

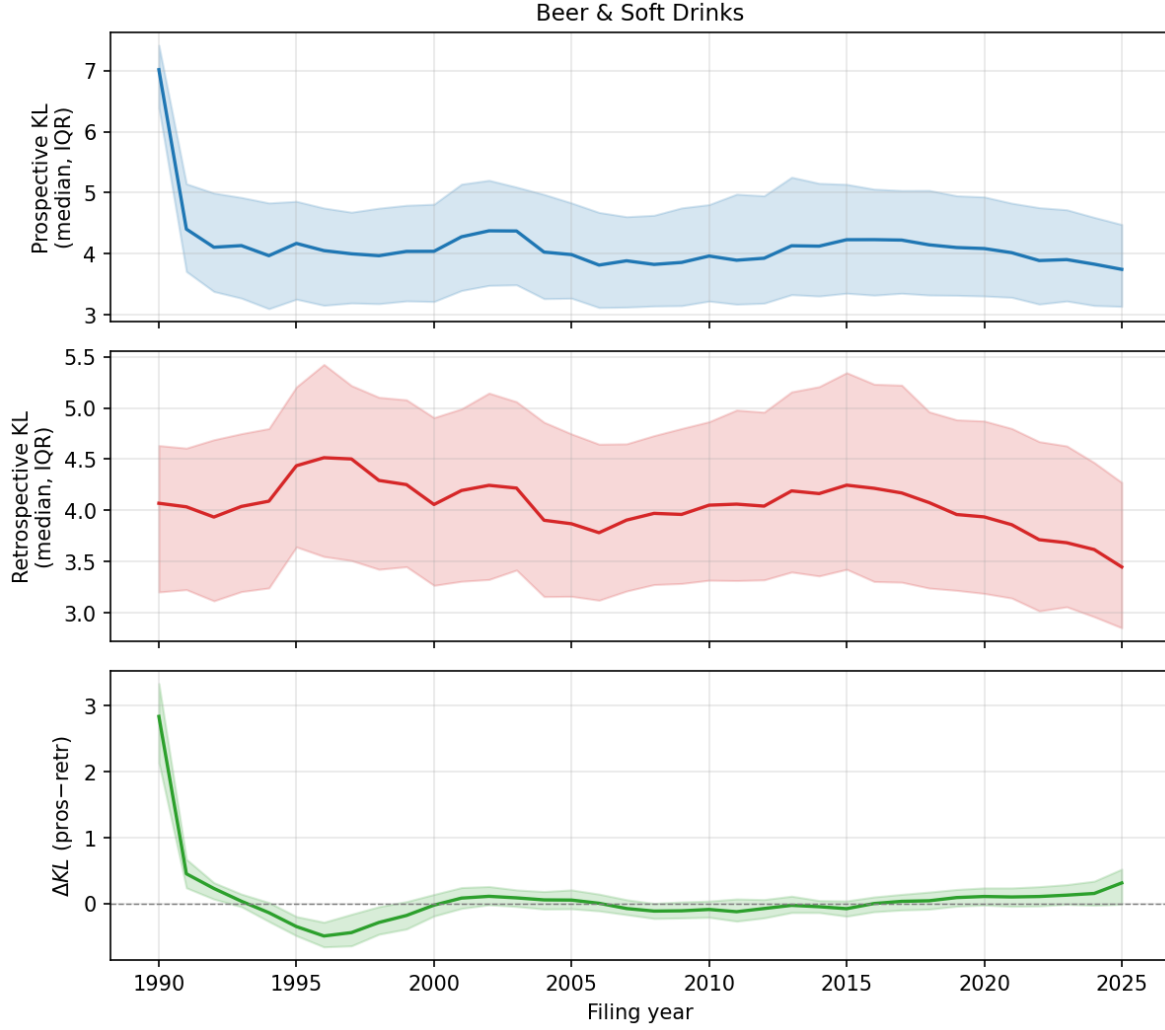


Figure 4: Beer & Soft Drinks: per-year median (line) and 25–75% range (band) of prospective KL, retrospective KL, and ΔKL .

where z_i is a z -scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i , and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

Industry	Outcome	Window	N	
Software & Electronics	Reached registration	any year	1,423,405	-0.1
Software & Electronics	Survived 5 yrs (registered & not cancelled at 5 y)	filed $\leq$ 2020	1,054,572	-0.0
Software & Electronics	Survived 10 yrs	filed $\leq$ 2015	732,789	-0.0
Software & Electronics	Currently live (Apr 2026)	any year	1,423,405	+0.1
Beer & Soft Drinks	Reached registration	any year	118,424	-0.1
Beer & Soft Drinks	Survived 5 yrs (registered & not cancelled at 5 y)	filed $\leq$ 2020	87,849	-0.0
Beer & Soft Drinks	Survived 10 yrs	filed $\leq$ 2015	59,807	-0.0
Beer & Soft Drinks	Currently live (Apr 2026)	any year	118,424	+0.1

The headline finding is the sign pattern. In Software & Electronics ($n = 1,146,415$), each one-standard-deviation increase in ΔKL *lowers* the log-odds of reaching registration by -0.042 (SE = 0.002) and lowers the log-odds of 5-year survival by -0.043 , but *raises* the log-odds of currently being live by $+0.038$. Decomposed jointly, prospective surprise penalizes every outcome and retrospective surprise reverses each sign, so the disadvantage at the registration gate is concentrated among filings whose future does *not* resemble them, while the long-run survival premium accrues to filings the future does come to resemble. Beer & Soft Drinks shows the same sign pattern.

Figures 5 and 6 display the per-bin registration-and-survival rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL . Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone. The currently-live indicator is dropped here; it is dominated by cohort effects (recent filings have not yet had time to fail) that the binned axis cannot absorb.

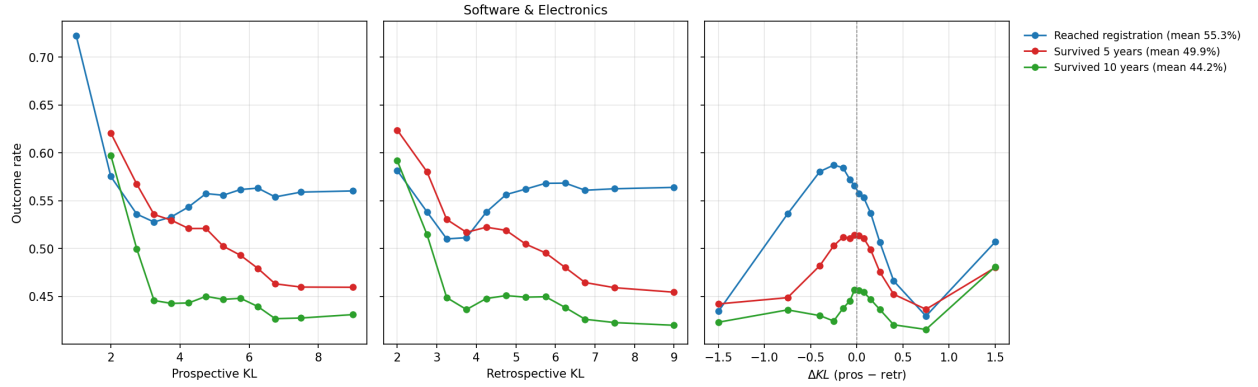


Figure 5: Software & Electronics: survival outcome rates by KL bin. The pattern across panels is consistent: high prospective KL penalizes registration and 5y survival; high retrospective KL is neutral-to-favorable; the difference ΔKL inherits a downward slope.

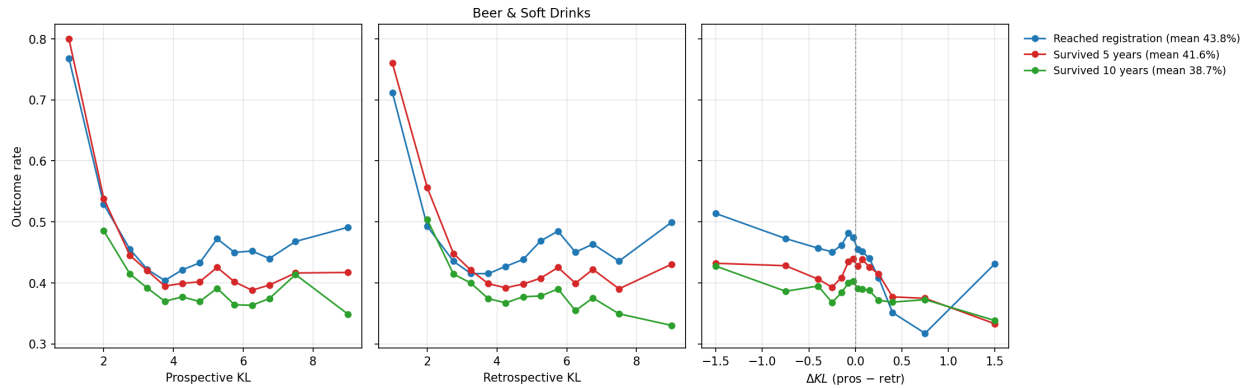


Figure 6: Beer & Soft Drinks: survival outcome rates by KL bin.

7 Outcome 2: Firm-year financials

To test whether innovation is rewarded financially we link USPTO owner names to SEC EDGAR firm names by aggressive normalization (uppercase, strip punctuation, drop legal-entity suffixes

such as INC, LLC, CORP, HOLDINGS, LIMITED, etc.) and exact match on the normalized key. The crosswalk recovers 4,651 USPTO owners covering 64,473 trademark filings (3.2% of the corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 4,110 firm-years across 1,408 unique CIKs over fiscal years 2009–2024.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm’s filing-level scores: $\overline{\Delta KL}_{c,y}$, $\overline{KL}^{\text{pros}}_{c,y}$, $\overline{KL}^{\text{retr}}_{c,y}$, the maximum within-window ΔKL , and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\Delta KL)_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

Outcome	Predictor(s)	N	First z-coef (SE)	Second z-coef (SE)	R^2
Gross margin	mean_dkl_3y	4,104	+0.029 (0.007)	—	0.016
Gross margin	mean_pros_3y, mean_retr_3y	4,104	+0.118 (0.043)	-0.170 (0.044)	0.044
Gross margin	max_dkl_3y, n_filings_3y	4,104	+0.027 (0.007)	+0.020 (0.008)	0.024
Net margin	mean_dkl_3y	3,742	-2.412 (2.018)	—	0.220
Net margin	mean_pros_3y, mean_retr_3y	3,742	-11.709 (11.304)	+14.924 (12.744)	0.220
Net margin	max_dkl_3y, n_filings_3y	3,742	-4.538 (3.342)	-2.893 (2.548)	0.221
Operating margin	mean_dkl_3y	3,768	-1.043 (0.664)	—	0.752
Operating margin	mean_pros_3y, mean_retr_3y	3,768	-6.106 (4.008)	+6.629 (4.240)	0.752
Operating margin	max_dkl_3y, n_filings_3y	3,768	-1.722 (1.006)	-0.316 (0.243)	0.754

The pattern from Section 6 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL raises gross margin by +2.9 percentage points (SE 0.7). Decomposed: prospective surprise raises gross margin by +11.8 pp/ σ (SE 4.3) and retrospective surprise lowers it by -17.0 pp/ σ (SE 4.3). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings the past resembles do not. R&D-intensity and log revenue, included as controls, have the expected signs but are not the focus here.

8 Discussion

The “pioneers vs. Blue Ocean” debate from Section 1 has both pieces empirically supported in this corpus. At the trademark gate, novelty is penalized: the most prospectively surprising filings are less likely to clear examination, less likely to file Section 8 maintenance, less likely to renew at 10 years. But conditional on survival, the same novelty axis is rewarded: marks the future imitates persist longer and are over-represented among currently-live registrations. At the firm-year level, public firms whose trademark portfolios sit in the prospectively-surprising half of the distribution earn higher gross margins than otherwise-comparable peers.

To make the underlying mechanics concrete, Table 1 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (**generative artificial**, **seltzer**, **cannabis**) that are common in the future corpus but rare in the look-behind

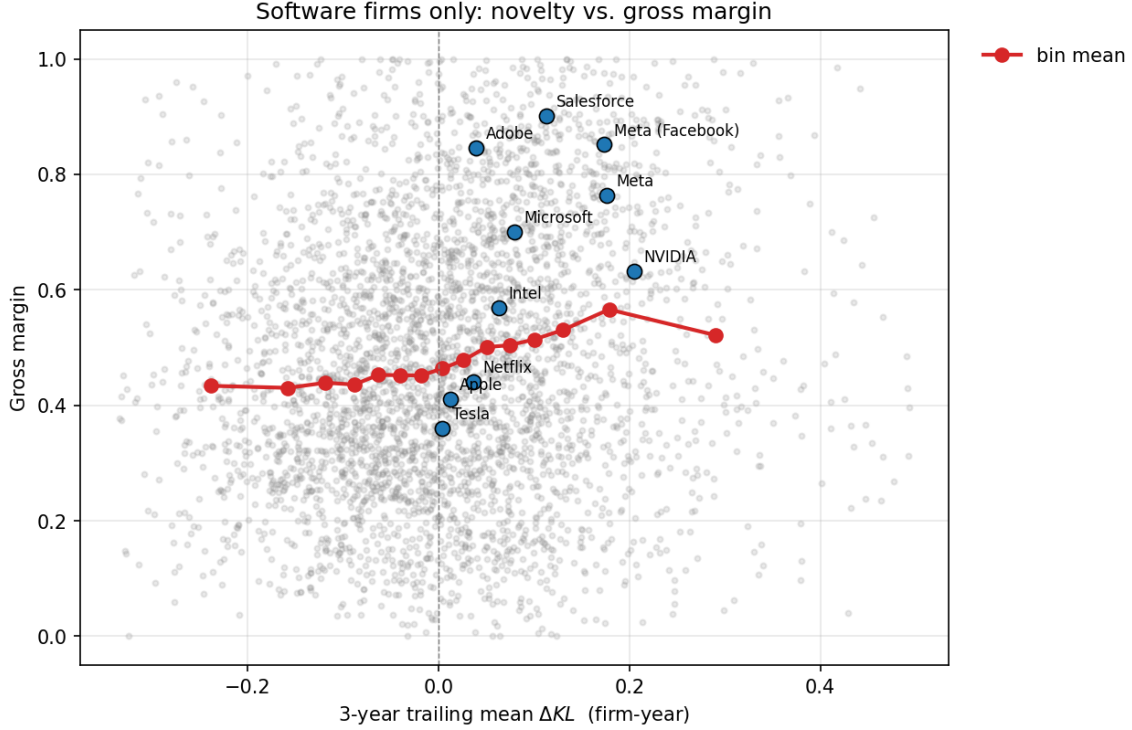


Figure 7: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL . Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL , gross margin clamped to $[0, 1]$). Red line: bin means. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (LIQUID DEATH), design novelty (AIRPODS casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Table 1: Selected filings with per-token KL contributions. Tokens are ordered by their contribution to each direction’s KL.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	software artificial, artificial intelligence, intelligence, artificial, software	software artificial, artificial intelligence, intelligence, artificial intelligence, software
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	model performing, document question-answering, tasks natural, generative text, performing generative	tasks, natural, model performing, question-answering, simulating, document question-answering
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	appearance enabling, modifying appearance, software modifying, enabling transmission, appearance	appearance enabling, modifying appearance, software modifying, enabling transmission, appearance
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	manipulating graphic, creating manipulating, images computer, graphic images, manipulating	manipulating graphic, creating manipulating, images computer, graphic images, program creating
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	servers secure, secure private, private networks, networks, private	servers secure, secure private, private networks, network private
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	sugar-based beer, sugar-based, seltzer, beer	sugar-based beer, sugar-based seltzer, beer
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	drink mixes, water nutritional, mixes concentrates, water sparkling, concentrates	drink mixes, mix concentrates, water nutritional, concentrated flavor enhanced
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	flavored brewed, brewed, brewed malt-based, malt-based alcoholic, nature beer	flavored brewed, brewed, brewed malt, nature beer, malt-based alcoholic
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	drinks, energy drinks, energy	drinks, energy drinks, energy

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim 1,400$ firms toward the $\sim 5,000$ ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high- ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc.*. $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC*. $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC*. $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.
Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.
Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC*. $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.
Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app...*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL*. $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.
Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc.*. $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.
Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer*
Alcoholic flavored brewed malt beverages, except beers

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC*. $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*